STAT 512 Project & Presentation Rubric

Grading Philosophy

This rubric emphasizes:

- Correct statistical inference over excessive methodology
- Clear reasoning over quantity of techniques
- Consistency between hypothesis, method, and conclusion

Rubric Breakdown

Category	Criteria	Points
1. Project Requirements & Cohesion (15 pts)		
Research Question Coverage	Number of research questions is no fewer than the number of group members. Each member contributes one primary question.	4
Variable Requirements	Response variable (Y) is continuous. At least one research question includes categorical predictor and appropriate continuous predictors, and studies the meaningful interaction term that is properly motivated and interpreted.	4
Coherence Across Questions	All research questions are logically connected and contribute to a single overarching research topic. Includes a final integrated conclusion or model.	5
2. Research Question, Variables, Design (15 pts)		
Research Question & Motivation	Clear, statistically testable question with meaningful context. Goes beyond descriptive comparison.	5
Variable Definition	Response and predictors clearly defined; correct types and justification of inclusion.	5

Model Framing	Correct regression structure (interaction, dummy variables); baseline interpretation explained.	3
Hypothesis Formulation	Hypotheses correctly written in parameter form (e.g., $\beta_j = 0$) and aligned with model and test.	4
3. Methodology & Model Building (25 pts)		
Model Building Logic	Clear pipeline: Exploratory Data Analysis $\rightarrow$ Model $\rightarrow$ Diagnostics $\rightarrow$ Decision $\rightarrow$ Final model. Each step justified.	8
Diagnostics Interpretation	Correct interpretation of: Residual vs fitted (linearity/variance), QQ plot (normality), VIF (multicollinearity).	9
Use of Remedies	Remedies are necessary, justified, and minimal. Avoids overuse of techniques.	8
4. Inference & Statistical Reasoning (20 pts)		
Hypothesis Testing Execution	Correct use of t-tests, F-tests, or linear hypothesis tests. Clear connection between hypothesis and test.	10
Interpretation of Results	Accurate statistical interpretation. Uses “associated with” instead of causal language. Results consistent with model.	10
5. Model Quality & Validation (15 pts)		
Model Selection	Model chosen based on appropriate criteria (Adjusted R^2, AIC, interpretability), not just R^2.	8
Validation & Generalization	Includes validation (e.g., k-fold) and discussion of limitations and generalizability.	7
6. Discussion & Conclusions (5 pts)		
Discussion	Conclusions clearly tied to research questions and hypotheses. Includes limitations and future improvements.	5
7. Presentation Quality (5 pts)		
Presentation	Slides are clear, concise, and well-organized. Communication is effective and balanced across team members.	5
Total		**100**